

CERTIFICATE PROGRAM IN AGENTIC AI





Listen-only mode



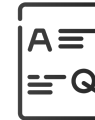
**Type your questions
in the Q&A box**



**Ask questions which are in the
interest of the larger audience**



**Recording & Slides will be
available 6 hours post session**



**Answers to all unanswered questions
will be shared in the coming weeks**

Agenda

1 AI Landscape

2 Program Overview

3 Learning Journey

4 Q&A



Faculty: Dr. Ian McCulloh



- Dr. Ian McCulloh leads the Artificial Intelligence portfolio for Lifelong Learning at Johns Hopkins University, with faculty roles in Computer Science and Public Health.
- His research combines AI, neuroscience, and human behavior to create scalable AI systems that improve access to products, services, and healthcare.
- Previously, he was Accenture's Chief Data Scientist, where he built and led a 1,200-strong federal AI practice delivering advanced AI solutions for the U.S. government.
- A retired U.S. Army Lieutenant Colonel, Dr. McCulloh founded the West Point Network Science Center and served as Chief Strategist for Information Warfare at CENTCOM.
- He holds a Ph.D. in Computer Science from Carnegie Mellon University and has authored several significant publications, including over 100 peer-reviewed papers.



Faculty: Dr. William Gray-Roncal



- Dr. William Gray-Roncal is a Principal Research Scientist at Johns Hopkins APL, leading interdisciplinary work in neuroscience, AI, machine learning, and healthcare, with over 20 years of engineering experience across diverse domains.
- His research focuses on brain network estimation at the synaptic scale and rapid scene analysis, blending science, engineering, and education.
- He has mentored nearly 500 students through initiatives like College Prep and CIRCUIT, supporting top STEM talent and fostering student-led research.
- Dr. Gray-Roncal is a leading expert in connectomics and has co-authored high-impact research, including a Nature paper mapping mouse visual cortex connectivity at nanoscale resolution.
- He holds advanced degrees in computer science and electrical engineering, maintains multiple academic appointments at Johns Hopkins, and lives in Maryland with his family.



Faculty: Dr. Shelby Wilson



- Dr. Shelby Wilson is a Senior Data Scientist at Johns Hopkins APL and a faculty member in the Engineering for Professionals program, with expertise in applied mathematics, mathematical biology, and data science.
- Her research spans computational epidemiology, dynamical systems, and machine learning, with a leadership role in the NSF APPEX Center focused on pandemic modeling and public health.
- She was named a Fellow of the Association for Women in Mathematics in 2022 for her scientific impact and commitment to mentoring and outreach.
- Dr. Wilson holds a Ph.D. in Applied Mathematics from the University of Maryland and dual B.S. degrees in Mathematics and Computer Science from Spelman College.
- Her career includes postdoctoral research in France, faculty roles at Morehouse College and UMD, and over a decade of experience solving critical scientific problems through mathematical modeling.



Faculties in the program



**Dr. Ian
McCulloh**



**Dr. William
Gray-Roncal**



**Dr. Shelby
Wilson**

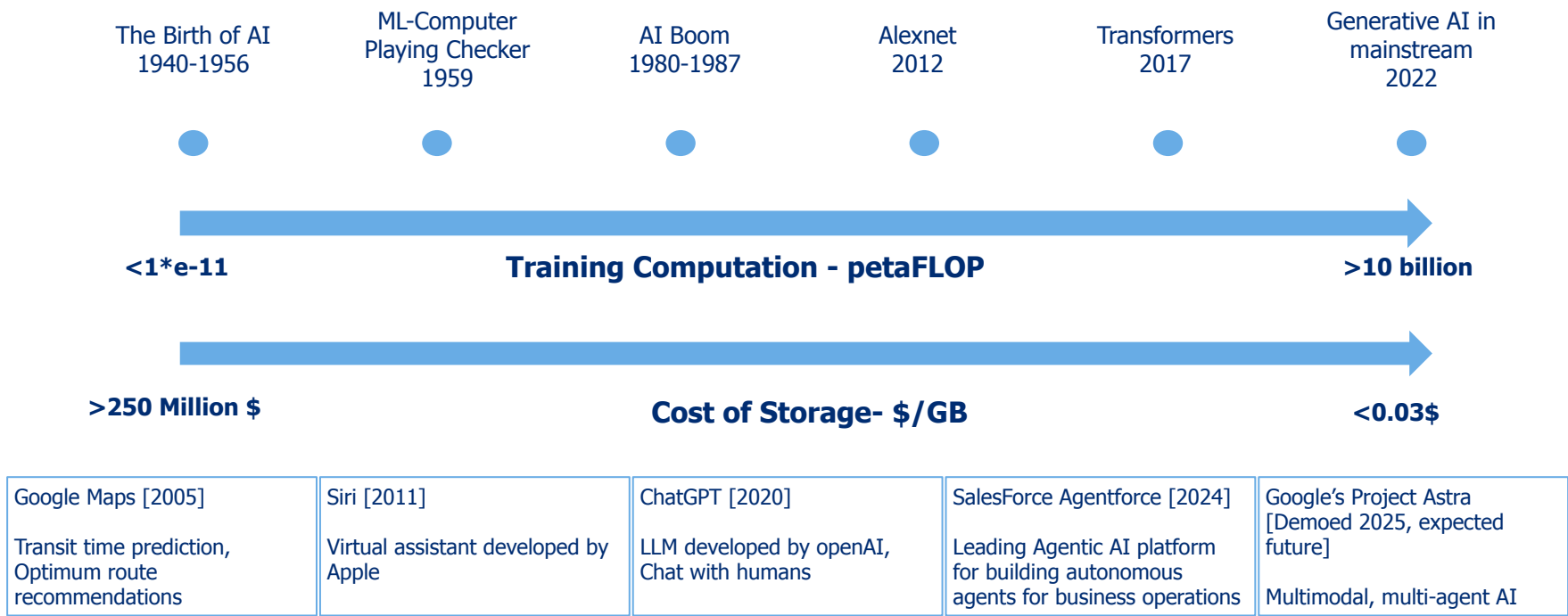


**Dr. Pedro
Rodriguez**



**Dr. Iain
Cruickshank**

AI has been here for a long time





AI is here to stay

Global AI race has already began. Choice is not whether we should use AI, rather the choice is about regulations, competition, and utility to benefit human society.

Regulation to address..

- Bias and Discrimination
- Ethical integrity
- Privacy and Data Protection
- Job Displacement
- Destructive use

More AI for a better world..

- Economic growth and innovation
- Evolution nature of human skills at work
- Better healthcare - diagnosis, treatment, drug discovery
- Breaking language barriers - world a global village
- Extreme personalisation



80% of AI initiatives fail, and generally for one or more of the three reasons

Incorrect use cases

- Selecting flashy use cases, not pragmatic ones
- Lack of clear business objectives
- Misalignment with business strategy

Poor data

- Insufficient data
- Bias in the data
- Data quality issues
- Data privacy and security concerns
- Inconsistent data labels

Poor communication

- Misaligned expectations
- Technologists lack domain expertise/experience
- Business leaders fail to engage/understand tradeoff decisions
- Cynicism and frustration



Learning Outcomes

1. Explain the fundamentals of Generative AI and LLMs for impactful business applications.
2. Implement Generative AI techniques like prompt engineering and RAG in alignment with business objectives.
3. Identify and mitigate ethical and risk-related challenges to ensure responsible AI implementation.
4. Describe the core components involved in building Agentic AI systems.
5. Design and develop end-to-end AI workflows and agents to improve efficiency in business functions.
6. Examine how Agentic AI leverages reinforcement learning and identify areas for improvement.
7. Evaluate human interaction in Agentic AI workflows and assess LLM-based agent performance.



Program Structure

1	Python Refresher for Agentic AI
2	Introduction to Large Language Models
3	Prompt Engineering Techniques & RAG
4	Project 1 Smart Data Processing Agent
5	Learning Break
6	Introduction to Agentic AI
7	Core Agent Concepts
8	Ethics, Safety, Alignment & Responsible AI

9	Planning & Reasoning Mechanisms
10	Project 2 Automated Research Agent
11	Learning Break
12	Multi-Agent Systems (MAS)
13	Reinforcement Learning in Agents
14	Interaction & Embodiment
15	Project 3 Customer Support Chatbot
16	Masterclass: The Future of Agentic AI



Frequently Asked Questions

What is this program about?

- The Certificate Program in Agentic AI is a 16-week online program designed to equip professionals with the skills to build autonomous, goal-driven AI agents capable of perceiving, reasoning, and acting independently. Learners will explore foundational theory alongside hands-on Python, projects to design intelligent systems that can plan, learn, and collaborate.

How is the program structured?

- You'll learn through recorded video content from Johns Hopkins University (JHU) faculty, complemented by live, interactive weekend Mentored Learning Sessions (MLS). Assessments are interspersed to reinforce learning.

How will I be assessed?

- You will be assessed through Projects and Graded Quizzes designed to evaluate your understanding and application of the concepts. To earn the certification, a cumulative score of at least 60% is required.

What tools do the learners need to install for the program before week 1?

- Before Week 1, learners are expected to install VS Code and Python. As an alternative, Google Colab can be used.



Frequently Asked Questions

What is the duration of pre-work and weekly pre-recorded content in hours?

- The duration of pre-work and weekly pre-recorded content will be ~2 hours

When will I get to know about my MLS timings and group?

- The PM will coordinate the MLS groups and session timings in advance of the sessions.

What support will I get?

- All sessions are recorded for flexible access. A dedicated PM for all your queries also you can raise support requests at any time, the responses is provided within 24 - 48 hours.



Q & A



Collaboration

Johns Hopkins

- **Academic Collaborator**
- Curriculum & content design
- Recorded Video lectures
- Case studies / Projects
- Certificate

Great Learning

- **Delivery Collaborator**
- Industry mentorship
- Learning management system
- Program Support
- Learning support



The Learning Philosophy

1. Learning is a hard process. It requires many different things to happen together. It takes a lot more than content to truly learn
2. For true learning to happen, one needs to
 1. Have a clear **learning path** that is **structured** well as well as **comprehensive**
 2. Have access to good **content** from **reputed, credible academicians**
 3. **Practice hands-on** by applying the learning on actual problems
 4. Have access to a **mentor** who can clarify doubts and provide industry perspective
 5. Be **tested frequently and get personalized feedback** to be aware of the progress
 6. Have **support** available during tough times
 7. Have access to **peers to network** with



Approach to Learning

Traditional Classroom

- Instructor prepares material to be delivered in class
- Learners listens to lecture in-class and other guided instructions and take notes
- Homework and assignments to demonstrate learning

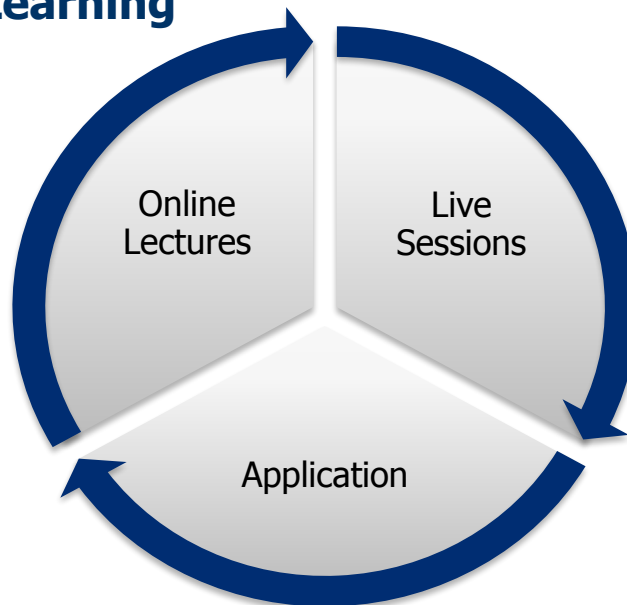
Flipped Classroom

- Instructor records and shares lectures outside of class
- Recorded Lectures and content to be watched before the session
- **In-class learning** includes applied learning and **higher order thinking tasks**
- Support through peers and Mentor



Vehicles of Delivering Learning

- Watch & Read
- ~2 hours of video lectures per week
- Reference reading material



- Mentor-led
- Online collaborative, interactive learning space
- Case studies
- Concepts
- Come prepared & ask questions

- Projects
- Quizzes

Weekly Learning Rhythm

Cohort	Module	Course (Week)	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Sep'25		1																																
		2																																
		3																																
		4																																

Illustrative

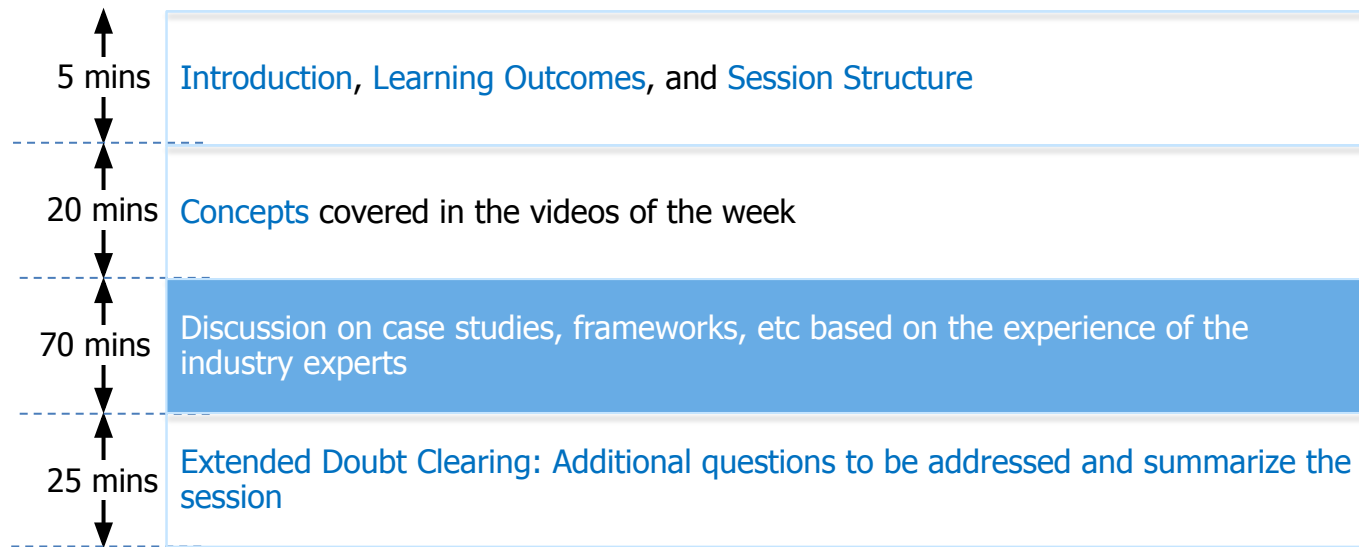
Video content, reading material & practice exercises

Live session

Deadline for assessment completion



Weekend mentorship session design





Assessments

In order to be eligible for the certificate, you will have to complete all mandatory courses with a minimum score of 60% in each.

Quizzes

- 1 quiz every learning week

Projects

- 3 hands-on projects

***Assessments in every course carry 90% weightage**

***Attendance of the Live sessions carries 10% weightage**

Learning Schedule



JOHNS HOPKINS

WHITING SCHOOL
of ENGINEERING

Executive and Professional Education

Certificate Program in Agentic AI

Learning Schedule - Sept 2025 Cohort



Great Learning

POWER AHEAD

Course #	Course	Program Week	Topic	Content Release	Live Session Weekend	JHU Master Classes	Assessment Deadline		
0	Pre-Work	0	Pre-work-01: Landscape of AI, Gen AI and Agentic AI	Available on enrollment			-		
			Pre-work-02: Python Basics	Available on enrollment			-		
			Pre-work Session 1- Business Applications of Agentic AI		7-Sept		-		
			Orientation Session		13-Sep		-		
0		0	Pre-work Session 2- Python Essentials for building Agentic AI solutions		14 Sept		-		
1	Prompt Engineering Foundations	1	Python Refresher for Agentic AI	11-Sep	20-Sep		21-Sep		
		2	Introduction to Large Language Models	18-Sep	27-Sep		28-Sep		
		3	Prompt Engineering Techniques & RAG	25-Sep	4-Oct	4-Oct	5-Oct		
		4	Project-1	25-Sep	11-Oct		26-Oct		
		5	Diwali Break						
2	Introduction to Agentic AI Design	6	Introduction to Agentic AI	23-Oct	1-Nov		2-Nov		
		7	Core Agent Concepts	30-Oct	8-Nov		9-Nov		
		8	Ethics, Safety, Alignment & Responsible AI	6-Nov	15-Nov	15-Nov	16-Nov		
		9	Planning & Reasoning Mechanisms	13-Nov	22-Nov		23-Nov		
3	Designing and Building Agentic Systems	10	Thanksgiving Break						
		11	Project-2	13-Nov	6-Dec	14-Dec	14-Dec		
		12	Multi-Agent Systems (MAS)	11-Dec	20-Dec		21-Dec		
4	Advanced Agentic AI		Christmas and New Year Break						
		13	Reinforcement Learning in Agents	1-Jan	10-Jan	11-Jan	11-Jan		
		14	Interaction & Embodiment	8-Jan	17-Jan		18-Jan		
		15	Project-3	8-Jan	24-Jan		1-Feb		
		16	The Future of Agentic AI	29-Jan	7-Feb		8-Feb		

1 Pre Work - Introductory material to help prepare foundations before the program launch and structure learning journey commences.

2 Program Orientation - Live session to launch the cohort and aims to orient learners to get the best out of the program ahead.

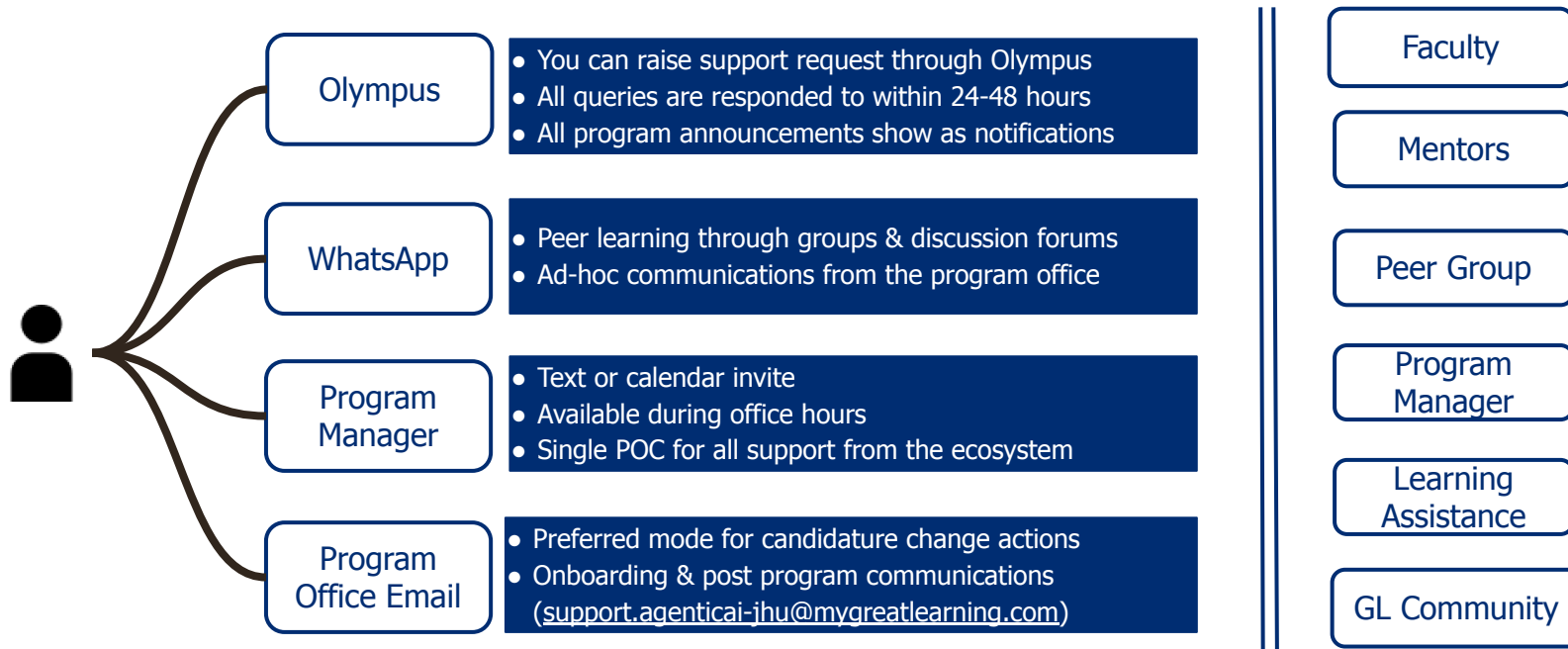
3 Live Session Weekend - Live learning sessions will happen either on a Saturday or a Sunday during weekends, the schedule slot will get finalized and communicated by the Program Office in the first week after the launch based on group formation, learners' and mentor's/faculty's schedules.

4 The assessment deadline is expected to be Sunday past midnight local time.

* This schedule is subject to change owing to a variety of reasons like design tweaks to improve learning experience and mentor availability changes. Program Office will ensure timely and clear communication in case of any changes



Learning Support Ecosystem





Make the most of the journey - Commit!

1. Consume all the content shared every week
2. Think about how to apply them in the real world
3. Prepare your questions before the sessions
4. Attend all the sessions
5. Participate during the sessions
6. Spend time working on assignments
7. Complete all the assignments
8. Reach out for support to enhance your learning journey
9. Ask questions
10. Share feedback with us

8-10 hours per week



Share feedback to help us improve

WHEN

Post session

- How did the session go?
- Any suggestions to add?

Mid Program

- How is the journey going?
- Discuss the progress so far

End Program

- How did your 16 weeks learning go?
- What went well and suggestions for improvement

HOW

Be Descriptive

Be Constructive

Share examples

Be realistic





Next Steps

- Login to Olympus at olympus.mygreatlearning.com
 - Complete your profile on Olympus (Name, linkedin profile, contact details)
 - Update your time zone on Olympus
- Go through the following courses:
 - Program Overview
 - Preparatory Course
- Join the first mentored learning session on weekend of September 20th, 2025 (*Link will be available under UPCOMING/ONGOING Activities on dashboard*)
- Submit the weekly quiz before the deadline, & in case of any questions, kindly raise a support request on Olympus through the 'Need Assistance' option



Q & A



Thank you

We'd love to hear your feedback!

Please share your feedback for the Orientation session

Wish you all the very best!

Please feel free to raise a support request through Olympus

Olympus Walkthrough

Thank you